



ARROW POWER!

VECTORS: DIRECTION & MAGNITUDE



THE LOST AUTO-RICKSHAW!

Rohan called out, "I'm 5 kilometres away!"
But WHERE exactly? North? South? East?
West?

Oh no — the auto-rickshaw driver is totally
LOST! 🚗 🧭

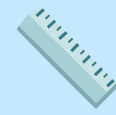
Distance alone is NOT enough.
We also need a DIRECTION!





Length

DIRECTION & MAGNITUDE?



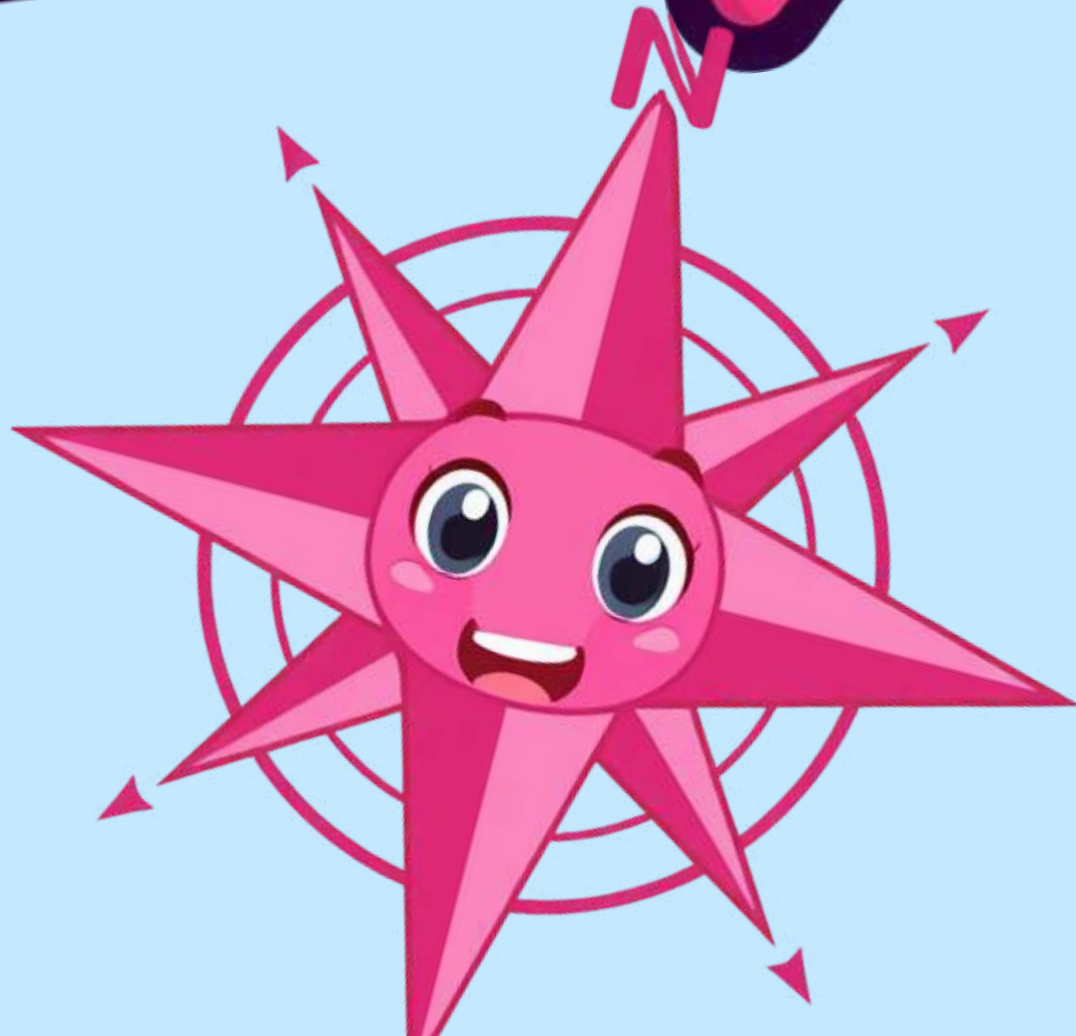
Magnitude = How FAR you travel (the distance!)



Direction = Which WAY you go!

Example: "5 kilometres towards Indiranagar Metro Station"

➡ Just saying '5 km' is not enough — you need BOTH to find the right place!



MEET THE MAGIC ARROW!

Vectors are magic arrows! Length shows strength, arrowhead shows the way.

- ◆ A SHORT arrow = small strength
 - ◆ A LONG arrow = big strength
 - ◆ The ARROWHEAD points the direction
- "Length gives the Strength,
Arrow shows the Way!"



KITE FLYING AT MAKAR SANKRANTI!



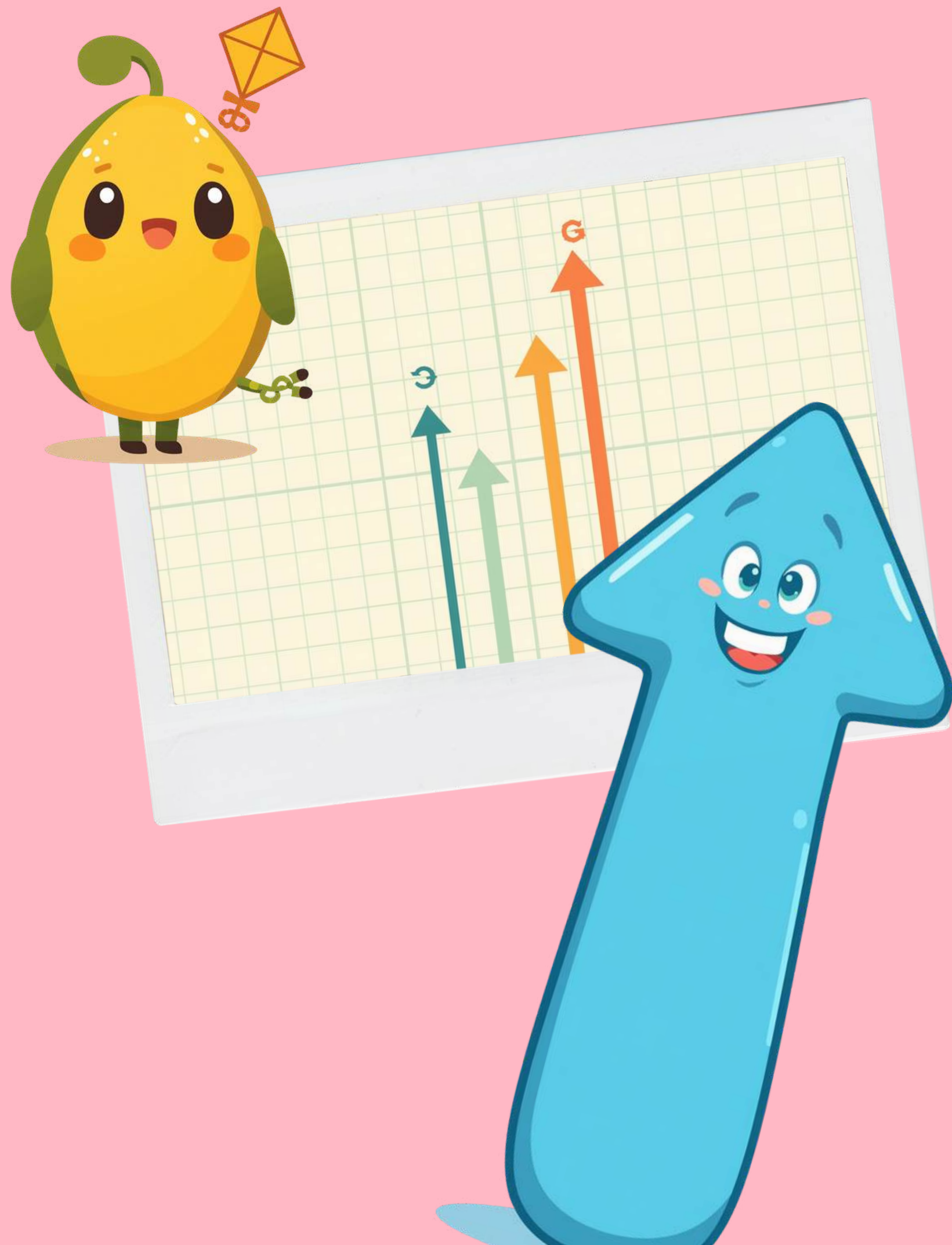
Short Arrow
Soft Breeze 🌿



Wind Blows North!
Direction Matters 🎯



Long Arrow
Strong Gust 🌪️



HOW ARROWS SHOW WIND POWER!

1

Short arrow = gentle wind (small magnitude). A tiny nudge of breeze barely moves the kite!

2

Long arrow = strong wind (large magnitude). A powerful gust sends the kite soaring high!

3

Arrow points North = direction of wind. The arrowhead always shows us which way the wind blows!

TUG-OF-WAR!

Team A pulls RIGHT with 5 units.

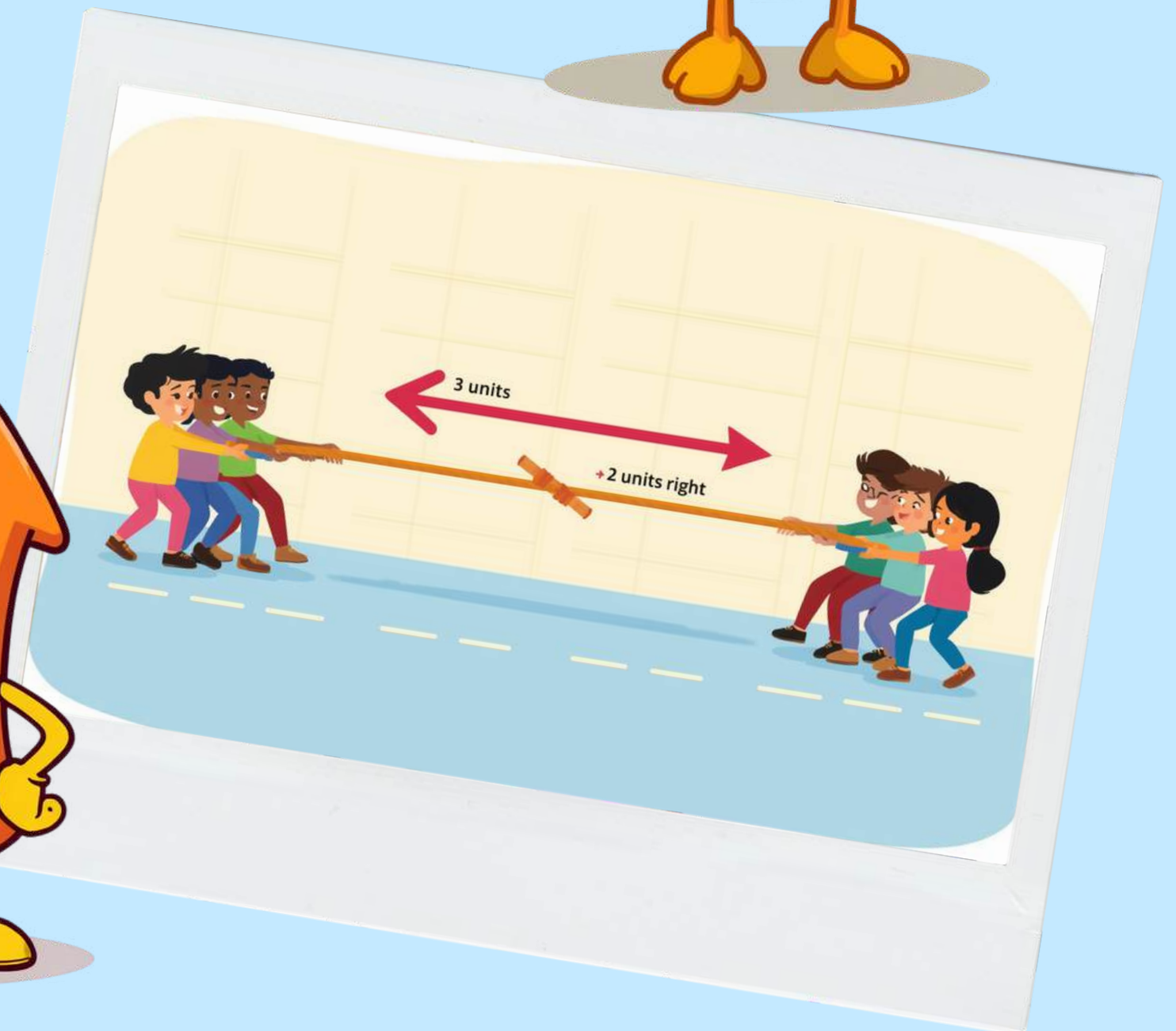
Team B pulls LEFT with 3 units.

Who wins? 🏆

When arrows pull opposite ways, subtract their lengths!

Result: $5 - 3 = 2$ units to the RIGHT!

Team A wins! ➡





TUG-OF-WAR EXPLAINED

Team A pulls RIGHT → 5 units

Team B pulls LEFT ← 3 units

When arrows pull opposite ways, subtract their lengths.

$$5 - 3 = 2 \text{ units RIGHT}$$

The stronger team wins — and the result arrow shows the direction of the winner!



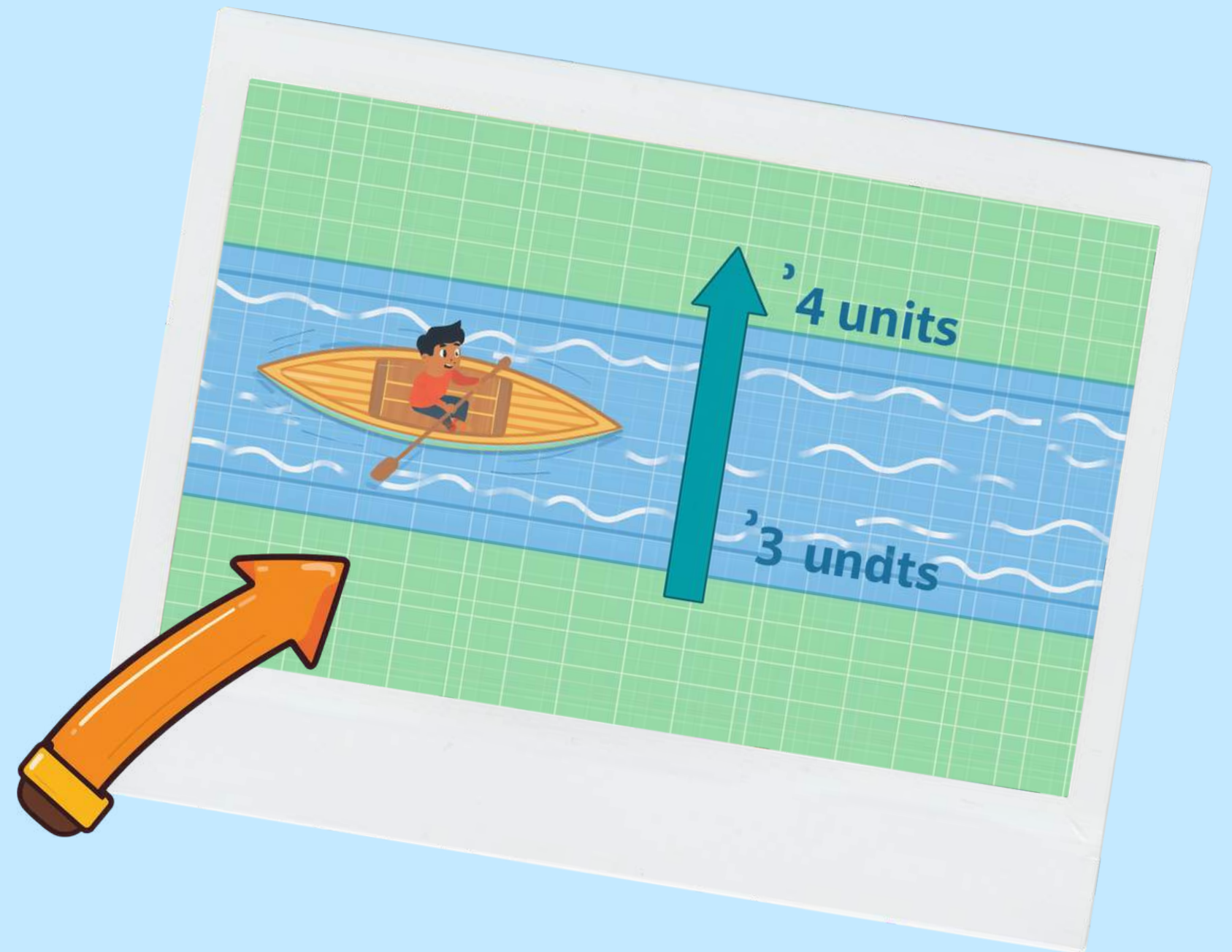
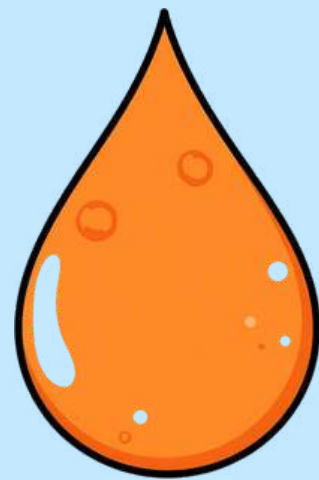
BOAT CHALLENGE – MOVING ACROSS THE RIVER!

Kabir rows his boat North with 4 units of force. But the river flows East with 3 units of push!

North arrow $\uparrow$ = 4 units (Kabir's rowing)

East arrow $\rightarrow$ = 3 units (river's flow)

Where does the boat actually go? 🚣



BOAT'S ACTUAL PATH!

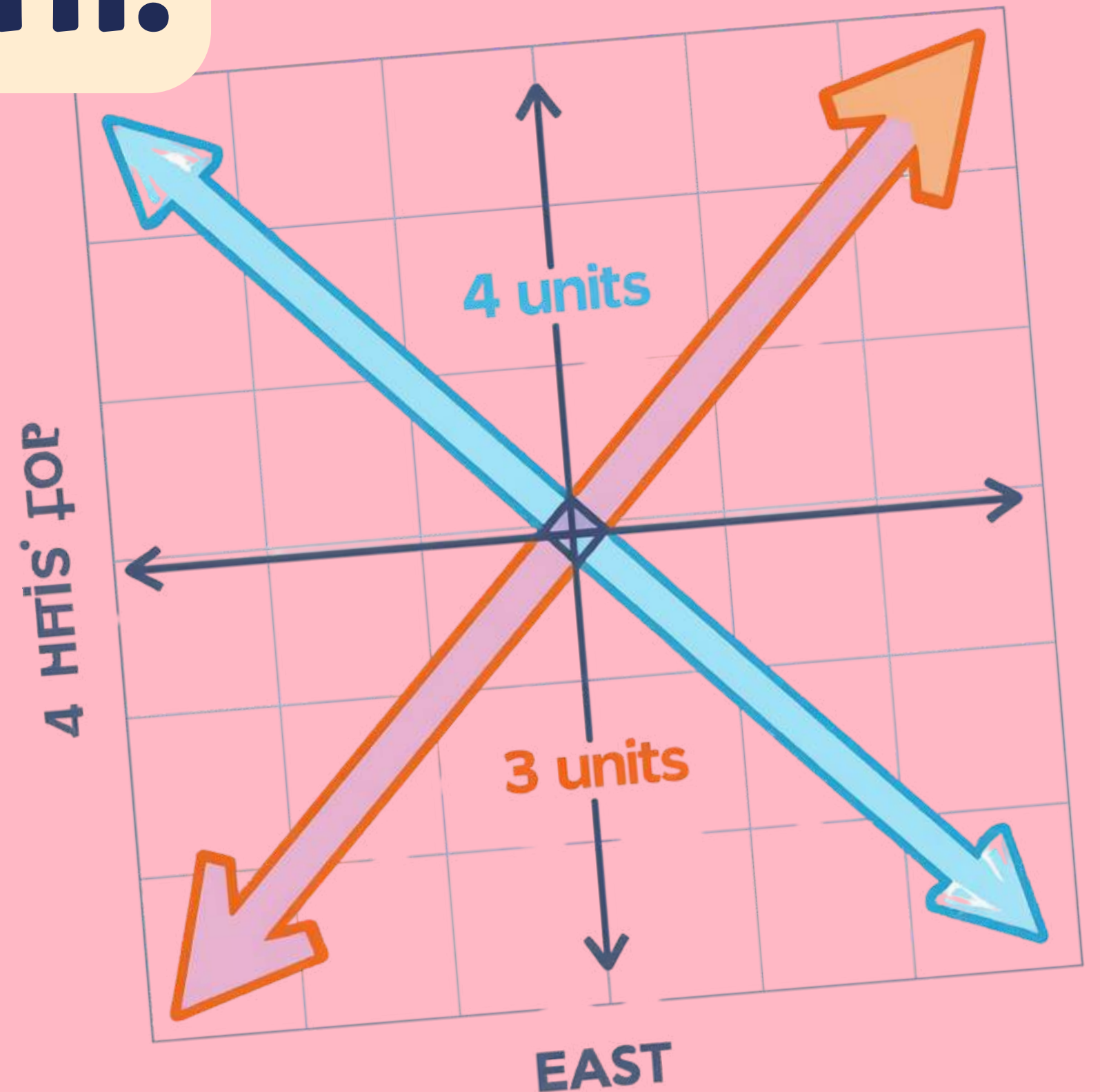
The boat moves diagonally where the two arrows combine!

➡ Kabir rows North: 4 units

➡ River flows East: 3 units

Result: The boat travels Northeast — combining both forces into one diagonal path!

"When two arrows join, a new path is born!"





LET'S PRACTICE TOGETHER!

1

Tug-of-War (Rassi Kheench!) — Pull the rope! Which team's arrow is longer? Who wins? Draw the arrows and find the winner!

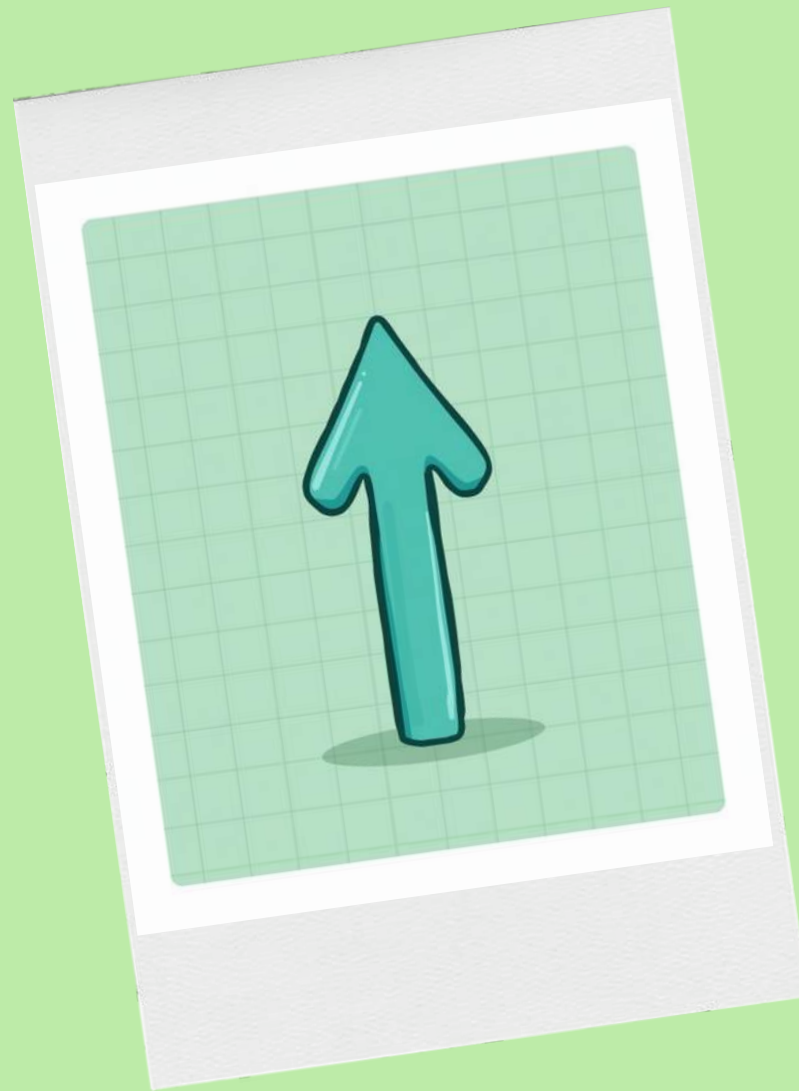
2

Boat Rowing on the Ganges — Kabir rows North, the river flows East. Draw both arrows on the grid and find where the boat actually goes!

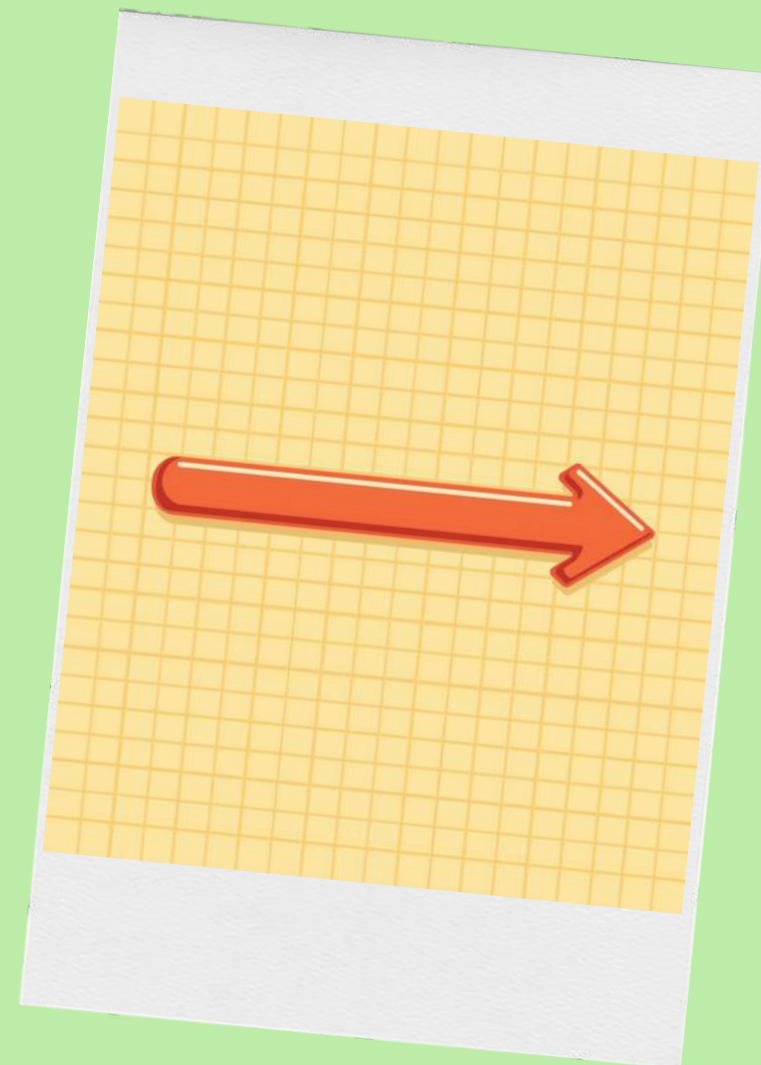
3

Guess the Arrow! — Look at the picture. Which direction is the wind blowing? Is the arrow short (gentle) or long (strong)? You decide!

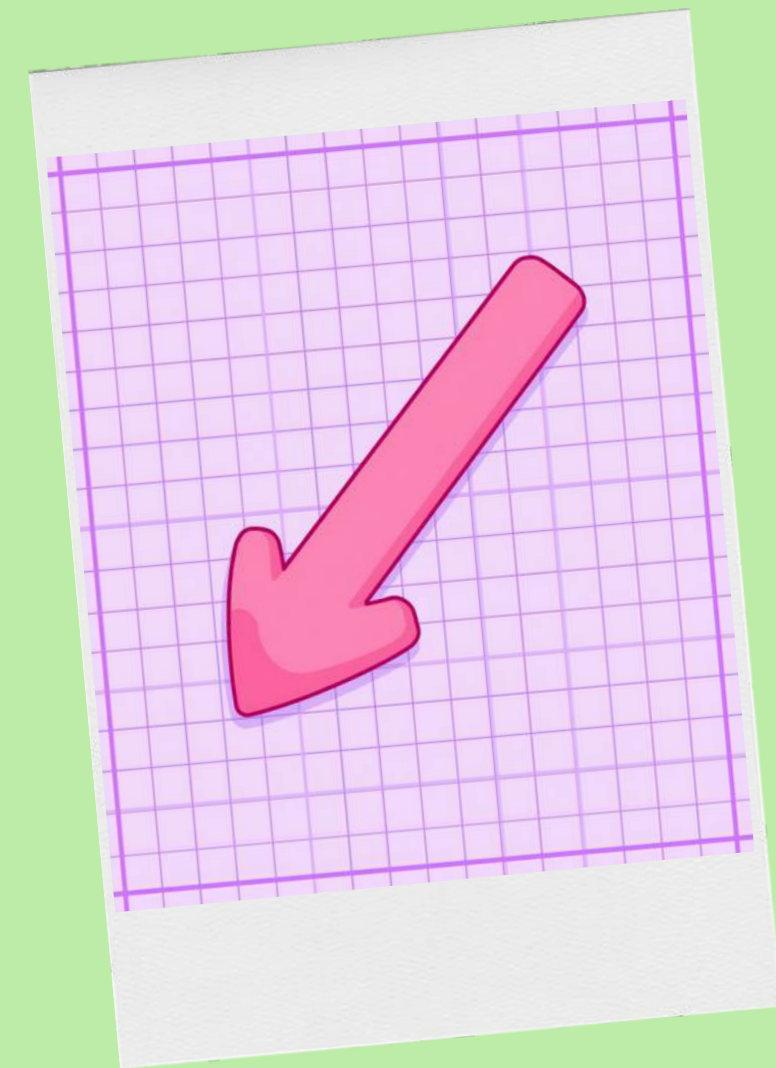
WHICH ARROW SHOWS A STRONG WIND BLOWING EAST?



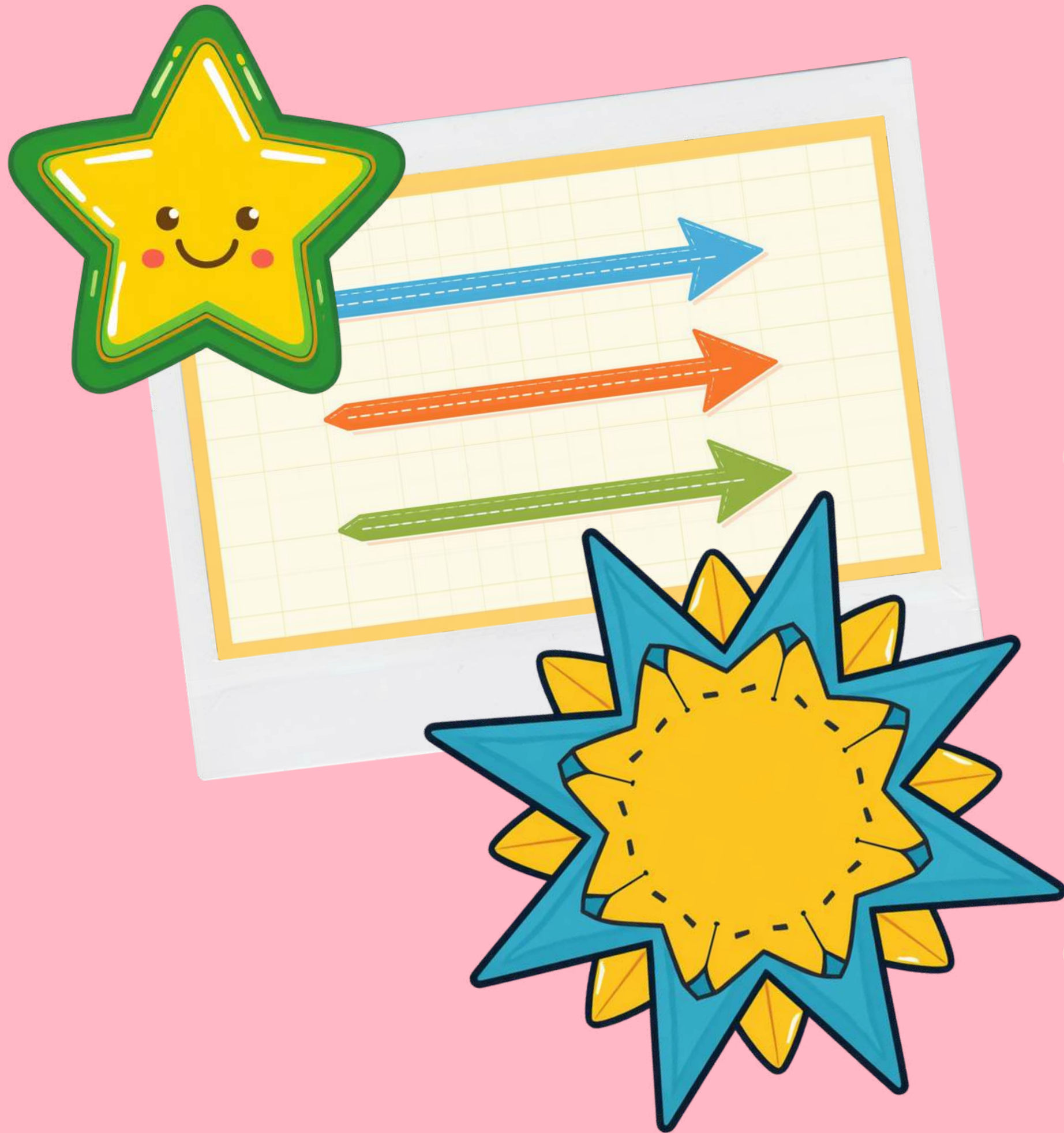
A) Short arrow —
North



B) Long arrow — East ✓



C) Medium arrow —
Southwest



QUICK RECAP – ARROW POWER!

1

Magnitude = length of the arrow. It tells us how strong or how far — a longer arrow means more power or greater distance!

2

Direction = where the arrow points. It shows us which way to go — North, South, East, West, or anywhere in between!

3

Adding arrows = combining strength and direction. When two vectors meet, we combine them to find the final path and power!



**THANKS FOR JOINING
ARROW POWER!
KEEP EXPLORING VECTORS IN YOUR WORLD.
"LENGTH GIVES THE STRENGTH,
ARROW SHOWS THE WAY!"**

